# CPSY 301 - Software Projects: Analysis, Design, and Management

**Description**
This course explores key project management concepts in software design and code management. Application of learning will be emphasized through team projects and collaboration.

3 Credits

**Time Guidelines**
The standard instructional time for this course is 45 hours.

**Prerequisites**

- One of:
  - CPSY 200
  - PROJ 304

**Course Assessment**

| | |
|---|---|
| Case Studies | 30% |
| Project | 70% |
| Total | 100% |

Use of Turnitin

The instructor may submit student work in this course to Turnitin's text-matching software program, to help assess the academic integrity of student work in this course. Turnitin results may be considered as one piece of evidence in academic misconduct hearings. Turnitin is an American company that keeps no unencrypted student identity data in the United States unless the students themselves choose to share this information in their submission. Students should limit their sharing of personally identifiable information by not including their names and student ID numbers within the text body of submitted assignments. SAIT has carefully reviewed this company's data management procedures.

**Other Course Information**
Blended (course site and F2F)

**Course Learning Outcomes**
1. Explain and Apply the Software Development Process

Objectives:

1.1 Distinguish between software development and software engineering in a project context

1.2 Compare project process models and select an appropriate model for a given project

1.3 Select and apply software project management methodologies (e.g., Agile, hybrid, predictive)

1.4 Apply CASE and project management tools to support team collaboration and documentation

2. Create and Validate a Requirements Model

Objectives:

2.1 Analyze, refine, and prioritize requirements derived from a project idea or client need

2.2 Develop and validate use case diagrams and detailed use case descriptions

2.3 Apply requirements analysis best practices to manage scope and change

2.4 Produce a validated stakeholder-reviewed requirements model suitable for design, estimation, and scheduling

3. Create Architectural, Component, System, and Database Design Models

Objectives:

3.1 Analyze and construct class diagrams to represent system structure, relationships, and responsibilities

3.2 Model and interpret system behaviour using sequence and activity diagrams to represent interactions and workflows

3.3 Analyze and model system states and transitions using state machine diagrams

3.4 Design and evaluate system architecture using component and deployment diagrams to represent system organization and execution environments

3.5 Analyze and design data models using Entity Relationship Diagrams (ERDs) to represent data requirements and relationships

3.6 Integrate and justify database, interface, and system components into a cohesive and consistent system design

3.7 Synthesize and produce a complete, consistent, and implementation-ready system and database design package

4. Create a Detailed Project Plan, Schedule, and Prototype

Objectives:

4.1 Demonstrate effective team collaboration, communication and professional working practices

4.2 Decompose project work into tasks and develop effort and duration estimates

4.3 Create a complete project schedule including milestones, dependencies, and timelines

4.4 Allocate resources and responsibilities to project tasks and milestones

4.5 Identify, analyze, and evaluate project risks and define mitigation strategies

4.6 Develop an interactive user interface prototype using tools such as Figma

4.7 Develop a system prototype demonstrating key system functionality.

**SAIT Policies and Procedures:**
Students should familiarize themselves with SAIT's Policies and Procedures, including but not limited to those in the Academic Student, Conduct, and Health, Safety, and Environment domains.

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